

Algebra I - MAT 1207

Evaluation Criteria :-

- * Tutorial and Assignments - 10%
 - * Attendance - 05%
 - * Mid Semester Exam - 25%
 - * End Semester Exam - 60%
- 100%

Ref :-

- A first Course in Abstract Algebra.
- J.B Foclegh.

Chapter :-

- Sets and Logics
- Relatives, functions and Binary Operations.
- Permutations.
- groups
- Introduction to number theory.

* Set theory

Defⁿ :-

Set is a collection of objects. These objects are called elements of the set.

Eg:- $A = \{\text{Number of students in the classroom}\}$

$$A = \{a_1, a_2, a_3, \dots, a_n\}$$

* Sets are finite or infinite.

Eg:- $[0, 1] \rightarrow \mathbb{R}$

This is a infinite set.

$\mathbb{N} = \{1, 2, 3, \dots\}$ - infinite set.

$\mathbb{R}$ - Set of Real numbers - infinite set.

$A = \{1, 2, 3, \dots, 10\}$ - finite set.

$\mathbb{Q} = \{p/q \mid p, q \in \mathbb{Z}, q \neq 0\}$ - rational numbers.
infinite set.

$\mathbb{P} = \{2, 3, 5, 7, 11, \dots\}$ - infinite set.

* Cardinality of the Set.

- The number of elements in the set is called Cardinality of the set.

Denoted by :- $|A|$, A - Set

Eg:- $A = \{1, 2, 3, \dots, 100\}$

$$|A| = 100$$

Eg:- $A = [0, 1]$

$$|A| = \alpha$$

* $\emptyset$ - Empty Set = $\{ \}$

Eg:- Find the elements of $A = \{x \mid x^2 \leq 20\}$, $x \in \mathbb{Z}$

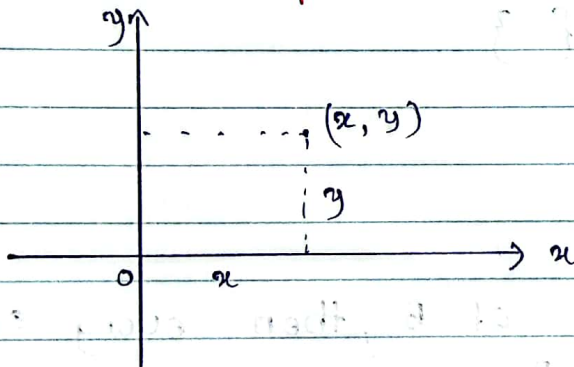
$$A = \{-4, -3, -2, -1, 0, 1, 2, 3, 4\}$$

$$|A| = 9$$

$$B = \{\emptyset, a, \{a, b\}, b, \{b\}\}$$

$$|B| = 5 \quad |\{a, b\}| = 2$$

* Cartesian products:-



$$A = \{a, b, c\}$$

$$B = \{d, e\}$$

$$A \times B = \{(a, d), (a, e), (b, d), (b, e), (c, d), (c, e)\}$$

$$A \times B = \{(x, y) \mid x \in A \text{ and } y \in B\}$$

* If $|A| = m$ and $|B| = n$

$$|A \times B| = mn$$

$$A \times B \times C = \{(a, b, c) \mid a \in A, b \in B, c \in C\}$$

Eg:- $A = \{a, b, c\}$

$$A^2 = A \times A = \{(a, b), (a, c), (b, a), (b, c), (c, a), (c, b), (a, a), (b, b), (c, c)\}$$

i.e. $|A^2| = |A|^2 = 3^2 = 9$

if $|A| = m$

$$|A^r| = |A|^r = m^r$$

$\emptyset$ - empty set

A - set $A \neq \emptyset$

$$\emptyset \times A = \emptyset$$

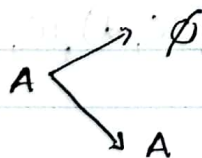
$$A = \{1, 2\}, \quad \emptyset = \{\}$$

$$|\emptyset \times A| = |\emptyset| = 0$$

* Subsets :-

- If A is a subset of B, then every elements in A must also in B.

Eg:- $\emptyset \subseteq \{1, 2, 3\}$



trivial
Subsets

$$\begin{cases} A \subseteq A \\ \emptyset \subseteq A \end{cases}$$

- * A Power Set of A, $\mathcal{P}(A)$ is the set containing all possible subsets of A.

Eg:- $A = \{a, b\}$

$$\mathcal{P}(A) = \{\emptyset, \{a, b\}, \{a\}, \{b\}\}$$

$$|\mathcal{P}(A)| = 4$$

Eg:- If $|A| = m$ then how many elements are in $|P(A)|$?
 $|P(A)| = 2^m$

$$* P(\emptyset) = \{ \emptyset \}, \quad |P(\emptyset)| = 1$$

$$* P(\{\emptyset\}) = \{ \emptyset, \{\emptyset\} \}, \quad \emptyset \subset \{\emptyset\}$$

$$\{\emptyset\} \subset \{\emptyset\}$$

Eg:- Let $|C| = k$ and $|D| = j$

$$|P(C \times D)| = 2^{kj}$$

$$|C \times D| = |C| \times |D|$$

$$= kj$$

Eg:- Find two sets A and B such that $A \in B$ and $A \subset B$

Solⁿ:- $\emptyset \subset B$ if $A = \emptyset$
 $A \subset B$

$$B = \{ \emptyset \}$$

$$A \in B$$

Eg:- $\{ 1 + (-1)^n \mid n \in \mathbb{N} \} = \{ 0, 2 \}$ is a finite set.

Eg:- If $A \subset B$ and $B \subset C$ then show that $A \subset C$

Solⁿ:-

$$\text{Let } x \in A$$

$$\text{Since } A \subset B$$

$$\Rightarrow x \in B$$

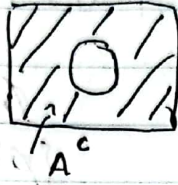
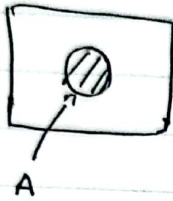
$$\text{Since } B \subset C$$

$$x \in B \Rightarrow x \in C$$

$$\Rightarrow A \subset C$$

* Complement of a Set :-

If A is a Set Complement is denoted as A' or $\bar{A}$ or A^c or A^c .



$$A \cap B = \{x \mid x \in A \text{ and } x \in B\}$$

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\}$$

$$A \setminus B = \{x \mid x \in A \text{ and } x \notin B\} \text{ (XOR) } = A - B$$

~~A = B~~

$$A = \{1, 3, 5, 7, 9\}$$

$$B = \{1, 4, 9, 11\}$$

$$A - B = \{3, 5, 7\}$$

Eg:- Show that if $A \subseteq B$ and $C \subseteq D$ then $A \cap C \subseteq B \cap D$

Let $x \in A \cap C$ we must show that $x \in B \cap D$.

$$x \in A \text{ and } x \in C$$

$$\text{Since } A \subseteq B$$

$$\Rightarrow x \in B \text{ and } x \in C$$

$$\text{Since, } C \subseteq D$$

$$\Rightarrow x \in D.$$

$$\Rightarrow x \in B \cap D$$

$$A \cap C \subseteq B \cap D$$

(ii) $A \cup C \subseteq B \cup D$ Let $x \in A \cup C$

$$\Rightarrow x \in A \quad \text{or} \quad x \in C$$

Since $A \subseteq B$

$$\Rightarrow x \in B$$

or

Since $C \subseteq D$

$$\Rightarrow x \in D$$

$$\therefore x \in B \cup D$$

$$\Rightarrow \underline{A \cup C \subseteq B \cup D}$$

Prove :-

ol. $A - (B \cap C) \subseteq (A - B) \cup (A - C)$

Solⁿ :-

Let $x \in (A - B \cap C)$

$$\Rightarrow x \in A \quad \text{and} \quad x \notin B \cap C$$

$$\text{and } (x \notin B \text{ or } x \notin C)$$

$$\Rightarrow (x \in A \text{ and } x \notin B) \text{ or } (x \in A \text{ and } x \notin C)$$

$$\Rightarrow x \in A - B \quad \text{or} \quad x \in A - C$$

$$\Rightarrow x \in (A - B) \cup (A - C)$$

$$\Rightarrow \underline{(A - B \cap C) \subseteq (A - B) \cup (A - C)}$$

H/w.

Prove :- 1. $\overline{A \cup B} = \overline{A} \cap \overline{B}$

$$A = B$$

2. If $A \subseteq B$ then $A - C \subseteq B - C$

$$A \subseteq B$$

$$B \subseteq A$$

Q2) IF $A \subseteq B$ then $A - C \subseteq B - C$

Let $x \in A - C$, we must show that $x \in B - C$

$$x \in A - C$$

$$x \in A \text{ and } x \notin C$$

Since $A \subseteq B$ therefore,

$$x \in B$$

$$\therefore x \in B \text{ and } x \notin C$$

$$\therefore x \in B - C$$

$$\therefore \underline{A - C \subseteq B - C}$$

Q3) $\overline{A \cup B} = \bar{A} \cap \bar{B}$

Let $x \in \overline{A \cup B}$, we must show that $x \in \bar{A} \cap \bar{B}$

$$x \notin A \text{ (or } x \notin B)$$

$$x \in \bar{A} \text{ and } x \in \bar{B}$$

$$x \in \bar{A} \cap \bar{B}$$

$$\therefore \overline{A \cup B} \subseteq \bar{A} \cap \bar{B}$$

Let $y \in \bar{A} \cap \bar{B}$, we must show that $y \in \overline{A \cup B}$

$$y \in \bar{A} \text{ and } y \in \bar{B}$$

$$y \notin A \text{ or } y \notin B$$

$$y \notin A \cup B$$

$$y \in \overline{A \cup B}$$

$$\therefore \bar{A} \cap \bar{B} \subseteq \overline{A \cup B}$$

$$\therefore \overline{A \cup B} \subseteq \bar{A} \cap \bar{B}$$

$$\bar{A} \cap \bar{B} \subseteq \overline{A \cup B}$$

$$\therefore \underline{\underline{\overline{A \cup B} = \bar{A} \cap \bar{B}}}$$

Eg:- $A \times (B \cup C) = (A \times B) \cup (A \times C)$ (by case I and case II)

$$A = \{1, 2\} \quad B = \{3, 5, 7\}$$

$$A \times B = \{(x, y) \mid x \in A, y \in B\}$$

$$A \times B \times C = \{(x, y, z) \mid x \in A, y \in B, z \in C\}$$

$$\mathbb{R} = \{\text{set of real numbers}\}$$

$$\mathbb{R} \times \mathbb{R} = \{(x, y) \mid x, y \in \mathbb{R}\}$$

$$\mathbb{R}^3 = \{(x, y, z) \mid x, y, z \in \mathbb{R}\}$$

$$\mathbb{R}^2 \times \mathbb{N} = \mathbb{R} \times \mathbb{R} \times \mathbb{N} = \{(x, y, z) \mid x, y \in \mathbb{R}, z \in \mathbb{N}\}$$

$$\mathbb{R} \times \mathbb{C} \times \mathbb{P} = \{(x, y, z) \mid x \in \mathbb{R}, y \in \mathbb{C}, z \in \mathbb{P}\}$$

Case - I To show that $A \times (B \cup C) \subseteq (A \times B) \cup (A \times C)$

Let $(x, y) \in A \times (B \cup C)$

$$\Rightarrow x \in A \quad \text{and} \quad y \in B \cup C$$

$$\Rightarrow x \in A \quad \text{and} \quad (y \in B \text{ or } y \in C)$$

$$\Rightarrow (x \in A \text{ and } y \in B) \text{ or } (x \in A \text{ and } y \in C)$$

$$\Rightarrow (x, y) \in A \times B \quad \text{or} \quad (x, y) \in A \times C$$

$$\Rightarrow (x, y) \in (A \times B) \cup (A \times C)$$

$$\Rightarrow \underline{A \times (B \cup C) \subseteq (A \times B) \cup (A \times C)}$$

Case II $(A \times B) \cup (A \times C) \subseteq A \times (B \cup C)$

Let $(x, y) \in (A \times B) \cup (A \times C)$

$$\Rightarrow (x, y) \in A \times B \quad \text{or} \quad (x, y) \in A \times C$$

$$\Rightarrow x \in A \text{ and } y \in B \quad \text{or} \quad x \in A \text{ and } y \in C$$

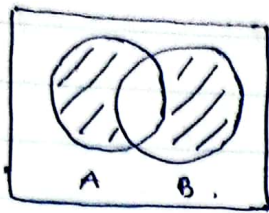
$$\Rightarrow x \in A \text{ and } (y \in B \text{ or } y \in C)$$

$$\Rightarrow x \in A \text{ and } y \in B \cup C$$

$$\Rightarrow (x, y) \in A \times (B \cup C)$$

$$\Rightarrow \underline{(A \times B) \cup (A \times C) \subseteq A \times (B \cup C)}$$

* Symmetric Difference of Sets.



$$(A-B) \cup (B-A)$$

$$A \Delta B = (A-B) \cup (B-A) \quad \left. \begin{array}{l} \\ \end{array} \right\} \text{XOR}$$

$$A \oplus B = (A-B) \cup (B-A)$$

Eg:- Show that, $(A \Delta B) \Delta B = A$

$$\text{Let } x \in (A \Delta B) \Delta B$$

$$\Leftrightarrow x \in (A \Delta B) \text{ XOR } x \in B$$

$$\Leftrightarrow \underbrace{x \in A \text{ XOR } x \in B}_{\text{XOR}} \text{ XOR } x \in B$$

$$\Leftrightarrow x \in A-B \text{ or } x \in B-A$$

$$\text{if } x \in A \Rightarrow x \notin B, \quad x \notin B$$

$$\therefore x \in A$$

$$(A-B) \cup (B-A) \quad \text{If } x \in B \Rightarrow x \notin A, \quad x \notin B$$

~~indexed~~

* Indexed Sets and well ordering principal.

$$A_1, A_2, A_3, \dots$$

$$\bigcap_{i=0}^n A_i$$

$$= A_0 \cap A_1 \cap \dots \cap A_n$$

$$\bigcup_{i=0}^n A_i$$

$$= A_0 \cup A_1 \cup \dots \cup A_n$$

* Well ordering principal :-

Any non empty subset of $\mathbb{N}$ has a least element

$$\mathbb{N} = \{1, 2, 3, \dots\}$$

Propositional Logics :-

Propositions :-

Declarative Sentences that can be true or false but not both at the same time.

Eg:- The result of adding two rational numbers is ~~an~~ irrational numbers. - False.

Eg:- Binary number 10 is used to represent "false" in logic. - True.

Eg:- Is this a ^{table} ~~proposition~~? - Not Proposition.

Eg:- Go out of the class! (command) - not proposition

Truth and falsity:-

propositions are either T or F we represent these with binary numbers 1 and 0.

T

F

1

0

on

off

We can combine simple proposition with different operators;

P, Q are two propositions

(Not) or and if P then Q

$\neg P$ $P \vee Q$ $P \wedge Q$ $P \rightarrow Q$

$\neg P$

if and only if

$P \leftrightarrow Q$

$\neg$ - negation,

* Syntax of Propositional Logic:

- well-formed formulas (wffs) are formed according to the rules of Propositional Logic.

Let P, Q, R are Propositions.

$$((\neg P) \wedge Q) \rightarrow R, (\neg P \wedge \neg Q) \wedge R$$

$$PQ \times$$

$$P \wedge \vee Q \times$$

* The main Connective in a formula is the one the (score) over the entire formula.

$$(\neg P \wedge Q) \rightarrow R$$

↑ main Connective

$$(\neg P \wedge \neg Q) \wedge (\neg R \wedge \neg S)$$

Calculating Truth Values.

P	$\neg P$
1	0
0	1

we can Calculate Complex well-formed formulas using truth tables. Truth tables list all the possibilities of truth for each set of simple propositions.

P, q, r three propositions :-

P	q	r
1	1	1
1	1	0
1	0	1
1	0	0
0	1	1
0	1	0
0	0	1
0	0	0

2^n if n propositions available
 $2^3 = 8$

P	TP
1	0
0	1

$$\text{Val}(TP) = 1 - \text{Val}(P) \wedge (P \leftrightarrow q)$$

Conjunctions :-

True iff both Conjunctions are true.

In English we write 'and'

* or

* but

* Although

* However

and $\wedge$

or $\vee$

P	q	$P \wedge q$	$P \vee q$
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	1

$$P \wedge q = \min(\text{Val}_p, \text{Val}_q)$$

$$P \vee q = \max(\text{Val}_p, \text{Val}_q)$$

$$\boxed{\text{Val}_p = \text{Value}_p}$$

Conditional :-

* if

* then

If P, q are Propositions.

P	q	$\neg P$	$\neg q$	$P \rightarrow q$	$\neg q \rightarrow \neg P$	$\neg P \rightarrow \neg q$
1	1	0	0	1	1	1
1	0	0	1	0	0	1
0	1	1	0	1	1	0
0	0	1	1	1	1	1

Biconditional :- P, q are propositions.

$$P \rightarrow q, q \rightarrow P \Leftrightarrow P \leftrightarrow q$$

$$(P \rightarrow q) \wedge (q \rightarrow P)$$

P	q	$P \rightarrow q$	$q \rightarrow P$	$P \leftrightarrow q$
1	1	1	1	1
1	0	0	1	0
0	1	1	0	0
0	0	1	1	1

$$\text{Ex: } \neg(P \wedge q) \vee (\neg P \wedge \neg q)$$

$$2. (q \rightarrow \neg q) \rightarrow (P \wedge \neg q) = P \quad (\text{Check using truth table})$$

Relations :-

Defⁿ :- A relation on a Set A is a Subset $R \subseteq A \times A$
 we often abbreviate the statement $(x, y) \in R$ as $x R y$
 The statement $(x, y) \notin R$ is abbreviated as $x \not R y$

eg:- $A = \{1, 2, 3, 4\}$

$$A \times A = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 1), (2, 2), (2, 3), (2, 4), (3, 1), (3, 2), (3, 3), (3, 4), (4, 1), (4, 2), (4, 3), (4, 4)\}$$

$$R = \{(1, 1), (2, 1), (2, 2), (3, 3), (3, 2), (3, 1), (4, 4), (4, 3)\}$$

is $R \subseteq A \times A$? Yes.

$$2 R 1, 2 \not R 4, 1 \not R 2.$$

Reflexive

Eg:- $R = \{(x, y) \in \mathbb{Z} \times \mathbb{Z} \mid x - y \in \mathbb{N}\}$

$$R \subseteq \mathbb{Z} \times \mathbb{Z} \quad (-1, 0) \Rightarrow x - y = -1 - 0 = -1$$

(but $-1 \notin \mathbb{N}$)

$$(x, x) \notin R \quad x - x = 0$$

$$0 \notin \mathbb{N}$$

$$\therefore (-1, 0) \notin R$$

$$R = \{(x, y) \in \mathbb{N} \times \mathbb{N} \mid x - y \in \mathbb{N}\}$$

$$(2, 0) \notin R$$

Types of Relations;

Defⁿ:- Suppose R is a relation on a Set A

01. Relation R is reflexive

if xRx $\forall x \in A$

i.e. $\forall x \in A, (x, x) \in R$.

02. Relation R is symmetric. if xRy implies yRx
 $\forall x, y \in A$

i.e. R is symmetry if $\forall x, y \in A$

$(x, y) \in R \Rightarrow (y, x) \in R$

03. Relation R is transitive if whenever xRy and yRz then xRz .

i.e. $\forall x, y, z \in A$, ~~xRy~~ $(x, y) \in R$ and $(y, z) \in R$
then $(x, z) \in R$.

$$R = \{(x, y) \in \mathbb{N} \times \mathbb{N} \mid x/y \in \mathbb{Q}\}$$

$$(x, y) \in R, (y, z) \in R$$

$$x/y \in \mathbb{Q}, y/z \in \mathbb{Q}$$

$$\frac{x}{y} \times \frac{y}{z} = \frac{x}{z} \in \mathbb{Q}$$

$$\therefore (x, z) \in R$$

Defⁿ:- A relation R on a Set A is an equivalence relation if it is reflexive, symmetric, transitive.

$$\text{eg:- } R = \{(x, y) \in \mathbb{N} \times \mathbb{N} \mid x/y \in \mathbb{Q}\}$$

is an equivalence relation.

No:

$$\cancel{n \mid x-y} = \frac{\cancel{n}}{\cancel{x-y}} = \frac{\cancel{n}}{\cancel{x-y}}$$

Eg: Let $n \in \mathbb{N}$, $n > 1$

$$R = \{ (x, y) \mid \cancel{x \equiv y} \in \mathbb{Z} \times \mathbb{Z} \mid x \equiv y \pmod{n} \}$$

$$n \mid x - y$$

R is defined on $\mathbb{Z}$

$$n \mid x - y = \frac{x - y}{n}$$

i) $\forall x \in \mathbb{Z}$, $(x, x) \in R$

$$x \equiv x \pmod{n} \quad n \mid x - x$$

$$n \mid 0$$

$\therefore R$ is reflexive.

ii) $\forall x, y \in \mathbb{Z}$

if $(x, y) \in R \Rightarrow x \equiv y \pmod{n}$

$$n \mid x - y$$

$$\Rightarrow n \mid [-(y - x)]$$

$$\Rightarrow n \mid [(-1)(y - x)] \quad n > 1$$

$$n \nmid (-1), \quad n \mid y - x$$

$$\Rightarrow y \equiv x \pmod{n}$$

$$\Rightarrow (y, x) \in R$$

$\therefore R$ is symmetric.

iii) $\forall y, x, z \in \mathbb{Z}$

if $(x, y) \in R$ and $(y, z) \in R$.

$$x \equiv y \pmod{n} \quad \text{and} \quad y \equiv z \pmod{n}$$

$$n \mid x - y$$

$$n \mid y - z$$

$$n \mid [(x - y) + (y - z)]$$

$$n \mid (x - z)$$

$$\Rightarrow x \equiv z \pmod{n}$$

$$(x, z) \in R$$

$\therefore R$ is transitive

$\Rightarrow R$ is equivalence relation.

Defⁿ :- Suppose R is an equivalence relation on A . Given element $a \in A$, the equivalence class containing a is the subset $\{x \in A \mid xRa\}$ of A consisting of all the elements of A that relate to a this is denoted by $[a]$.

$$A = \{-1, 1, 2, 3, 4\}$$

$$[a] = \{x \in A \mid xRa\}$$

Eg:- $R_1 = \{(-1, -1), (1, 1), (2, 2), (3, 3), (4, 4)\}$

$$[2] = \{x \in A \mid xR_1 2\} = \{2\}$$

$$[-1] = \{-1\}$$

$$[1] = \{1\}$$

$$[3] = \{3\}$$

$$[4] = \{4\}$$

$$[2] = \{2\}$$

$$[3] = \{3\}$$

$$[4] = \{4\}$$

$$[2] = \{2\}$$

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$$[2] = \{2\}$$

$$[3] = \{3\}$$

$$[4] = \{4\}$$

2. $R_2 = \{(-1, -1), (1, 1), (2, 2), (3, 3), (4, 4), (-1, 1), (1, -1), (-1, 3), (3, -1), (1, 3), (3, 1), (2, 4), (4, 2)\}$

R_2 - equivalence relation.

$$[-1] = \{x \in A \mid xR_2 -1\} = \{-1, 1, 3\}$$

$$[1] = \{-1, 1, 3\}$$

$$[2] = \{2, 4\}$$

$$[3] = \{3, -1, 1\}$$

$$[4] = \{4, 2\}$$

$$[-1] = [1] = [3] = \{-1, 1, 3\}$$

$$[2] = [4] = \{2, 4\}$$

Two distinct classes.

Eg:- Let P be the set of all polynomials with real Coefficients.

$$P = \{a_n x^n + a_{n-1} x^{n-1} + \dots + a_0 \mid n \in \mathbb{N}, \cup \{0\} \\ a_n, a_{n-1}, \dots, a_0 \in \mathbb{R}, a_n \neq 0\}$$

Define a relation R on P as follows given $f(x), g(x) \in P$, $f(x) R g(x)$, means that $f(x)$ and $g(x)$ have the same degree.

$$\text{i.e. } f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_0 \quad \left\{ \begin{array}{l} a_n \neq 0, a_n, \dots, a_0 \in \mathbb{R} \end{array} \right.$$

$$g(x) = b_n x^n + b_{n-1} x^{n-1} + \dots + b_0 \quad \left\{ \begin{array}{l} b_n \neq 0, b_n, \dots, b_0 \in \mathbb{R} \end{array} \right.$$

$$\begin{array}{ll} * \text{ Let } f(x) = x^3 + x + 1 & \times f(x) = \frac{1}{3}x^2 - 1 \\ g(x) = x^3 - 1 & g(x) = \sqrt{2}x^2 + 1 \\ (x^3 + x + 1) R (x^3 - 1) & (\frac{1}{3}x^2 - 1) R (\sqrt{2}x^2 + 1) \end{array}$$

$$\times \quad \left. \begin{array}{l} f(x) = x + 1 \\ g(x) = x^2 - 1 \end{array} \right\} f(x) \not R g(x).$$

$$\begin{aligned} [x^3 + 1] &= \{a_3 x^3 + a_2 x^2 + a_1 x + a_0 \mid a_3 \neq 0, a_3, a_2, a_1, a_0 \in \mathbb{R}\} \\ [a] &= \{x \in A \mid x R a\} \\ x^3 + x^2 + 1 &\in [x^3 + 1] \\ x^3 &\in [x^3 + 1] \end{aligned}$$

Eg:- $R = \{(x, y) \mid x \equiv y \pmod{n}, n \in \mathbb{N}\} \subseteq \mathbb{Z} \times \mathbb{Z}$

Suppose $n = 3$

$$R = \{x, y \mid x \equiv y \pmod{3}\}$$

$$[0] = \{x \in \mathbb{Z} \mid x R 0, x \equiv 0 \pmod{3}\}$$

$$3 \mid x - 0 \Rightarrow 3 \mid x$$

$$[0] = \{\dots, -15, -12, -9, -6, -3, 0, 3, 6, 9, \dots\}$$

$$[3] = \left\{ x \in \mathbb{Z} \mid x \equiv 3 \pmod{3}, \begin{array}{l} 3 \mid x-3 \end{array} \right\}$$

$$[3] = \{ \dots, -15, -12, -9, -6, -3, 0, 3, 6, 9, \dots \}$$

$$[0] = [3] = [6] = [9] = \dots$$

$$[1] = \left\{ x \in \mathbb{Z} \mid x \equiv 1 \pmod{3}, \begin{array}{l} 3 \mid x-1 \end{array} \right\}$$

$$= \{ \dots, -5, -2, 1, 4, 7, 10, \dots \}$$

$$[4] = \left\{ x \in \mathbb{Z} \mid x \equiv 4 \pmod{3}, \begin{array}{l} 3 \mid x-4 \end{array} \right\}$$

$$= \{ \dots, -8, -5, -2, 1, 4, 7, 10, \dots \}$$

$$[1] = [4] = [7] = [10] = \dots$$

$$[2] = \left\{ x \in \mathbb{Z} \mid x \equiv 2 \pmod{3}, \begin{array}{l} 3 \mid x-2 \end{array} \right\}$$

$$[2] = \{ \dots, -7, -4, -1, 2, 5, \dots \}$$

$$[5] = [2]$$

$$[8] = [5] = [11] = \dots$$

$$[0], [1], [2]$$

$$[0] \cap [1] \cap [2] = \emptyset$$

$$n=5$$

$$[0], [1], [2], [3], [4]$$

Suppose n

$[0], [1], \dots, [n-1]$ distinct equivalence classes

$$[0] \cap [1] \cap \dots \cap [n-1] \neq \emptyset$$

**** Th^m :-**

Suppose R is an equivalence relation on a set A . Suppose also that $a, b \in A$. Then $[a] = [b]$ iff aRb

Pf :-

Suppose $[a] = [b]$

we must show that aRb

Since R is equivalence relation
 R is reflexive.

$\forall a \in A \quad aRa$

$$[a] = \{x \in A \mid xRa\}, [b] = \{x \in A \mid xRb\}$$

$$a \in [a] = [b]$$

$$\Rightarrow a \in [b]$$

$$\Rightarrow aRb$$

Conversely, Suppose aRb

we must show that $[a] = [b]$

So first we show that

$$[a] \subseteq [b]$$

Let $c \in [a]$

cRa given that aRb

Since R is transitive (equivalence relation)

$$\Rightarrow cRb$$

$$\Rightarrow c \in [b]$$

$$\Rightarrow [a] \subseteq [b] \text{ --- ①}$$

//^{1st} Let $d \in [b]$

dRb given that aRb

Since R is symmetric (equivalence relation)

$$aRb \Rightarrow bRa$$

Since R is transitive

$$dRb \text{ and } bRa$$

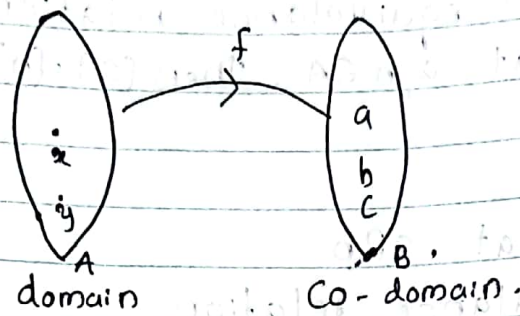
$$\Rightarrow dRa$$

$$d \in [a]$$

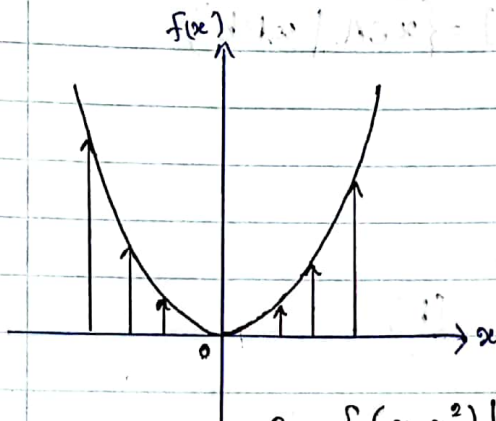
$$\Rightarrow [b] \subseteq [a] \text{ --- ②}$$

$$\text{①, ②} \Rightarrow [a] = [b]$$

Functions :-



Consider $f(x) = x^2$



$$R = \{(x, x^2) \mid x \in \mathbb{R}\} \subseteq \mathbb{R} \times \mathbb{R}$$

Defⁿ :-

Suppose A and B are sets. A function f from A to B ($f; A \rightarrow B$) is a relation $f \subseteq A \times B$ from A to B , satisfying the property that for each $a \in A$ the relation f contains exactly one ordered pair of the form (a, b) .

The statement $(a, b) \in f$, abbreviated as,

$$f(a) = b \quad f; A \rightarrow B.$$

$$(a, b) \in f \quad (a, c) \in f \Rightarrow (a, c) \notin f \text{ if } b \neq c.$$

$$f = \{(n, |n|) \mid n \in \mathbb{Z}\} \subseteq \mathbb{Z} \times \mathbb{N}$$

$$f(n) = |n|$$

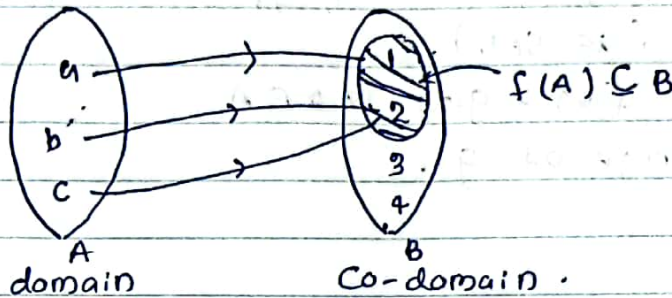
This is a relation from $\mathbb{Z}$ to $\mathbb{N}$.

Defⁿ:-

For a function $f: A \rightarrow B$

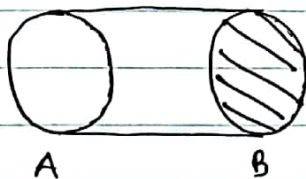
A - domain

B - Co-domain

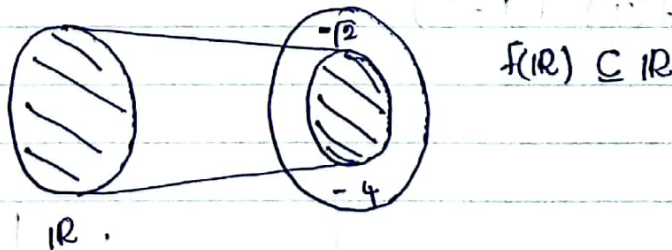


$f(A)$ - range of A

$$f(A) = \{y \in B \mid f(x) = y, x \in A\}$$



Eg:- $f; \mathbb{R} \rightarrow \mathbb{R}$
 $f(x) = x^2$



Eg:- $\phi; \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}$

$$\phi(m, n) = 6m - 9n = 3(2m - 3n) = 3k \quad k \in \mathbb{Z}$$

$$\phi = \{ (m, n), (6m - 9n) \mid (m, n) \in \mathbb{Z} \times \mathbb{Z}, (6m - 9n) \in \mathbb{Z} \}$$

$$= \{ (m, n), 3k \mid (m, n) \in \mathbb{Z} \times \mathbb{Z}, k \in \mathbb{Z} \}$$

Defⁿ :-

Two functions $f; A \rightarrow B$ and $g; A \rightarrow D$ are equal if $f = g$ (as set).

i.e. $f = g$ iff $f(x) = g(x) \quad \forall x \in A$

range of $f =$ range of g .

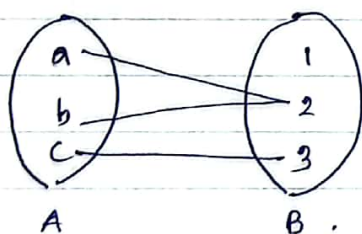
Defⁿ :-

A function $f; A \rightarrow B$ is

1. injective (1-1, one to one)

if $\forall x, y \in A, x \neq y$ implies that $f(x) \neq f(y)$

or $\forall x, y \in A$, if $f(x) = f(y)$ implies $x = y$



$$a \neq b$$

$$f(a) = f(b)$$

not 1-1

$$x = y \iff f(x) = f(y) \quad (1-1)$$

Eg:- $f(x) = x^2$

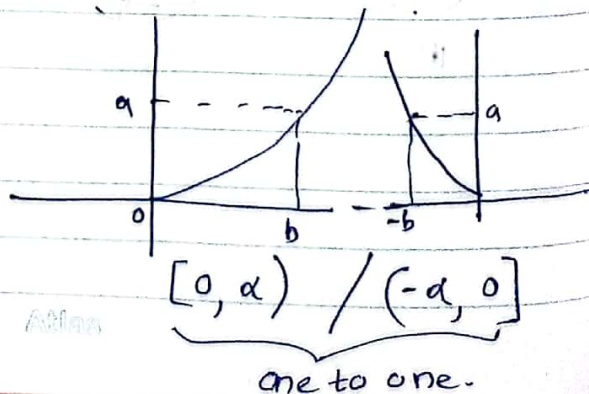
$$f; \mathbb{R} \rightarrow \mathbb{R}$$

$$x, y \in \mathbb{R}$$

$$f(y) = f(x)$$

$$y^2 = x^2$$

$$y = \pm x$$

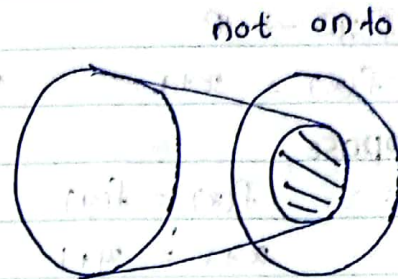
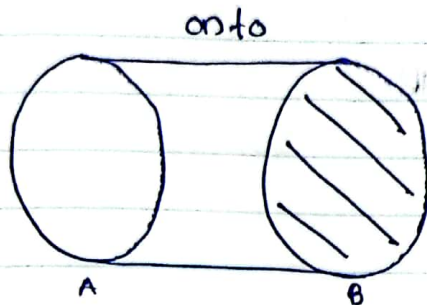


2. Surjective (onto)

if $\forall y \in B \exists \text{ an } x \in A$

Such that $f(x) = y$

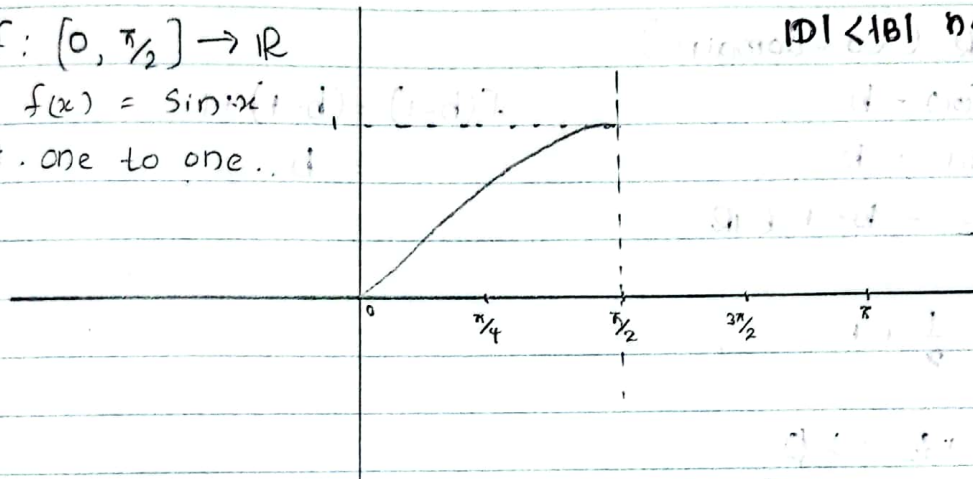
i.e. Range of $f = \text{Co domain}$.



$$f: [0, \pi/2] \rightarrow \mathbb{R}$$

$$f(x) = \sin x$$

* one to one.



$|D| \geq |B|$ may be surjective

$|D| < |B|$ not surjective.

$$f: \mathbb{R} \rightarrow [0, \infty)$$

$$f(x) = x^2$$

Let $x, y \in \mathbb{R}$

$$\text{if } f(x) = f(y)$$

$$x^2 = y^2$$

$$x = \pm y$$

* not 1-1

$$x, y \in [0, \infty)$$

$$f(x) = y$$

$$x^2 = y$$

$$x = \pm \sqrt{y}$$

$$y = 2 \quad \sqrt{y} = \sqrt{2}$$

$$f(\sqrt{2}) = 2$$

$$f(-\sqrt{2}) = 2$$

* onto.

03). Bijective :-

If f is both injective and surjective then f is bijective.

Eg:- $f: \mathbb{R} \rightarrow \mathbb{R}$

$$f(x) = x+1$$

$\forall x, y \in \mathbb{R}$

Suppose;

$$f(x) = f(y)$$

$$x+1 = y+1$$

$$x = y$$

$\therefore$ one to one.

$\forall b \in \mathbb{R}$ (co-domain)

$$\text{If } f(x) = b$$

$$x+1 = b$$

$$x = b-1 \in \mathbb{R}$$

$$f(b-1) = (b-1)+1 = b.$$

Eg:- $f(x) = \frac{1}{x} + 1$

Eg:- $h: \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Q}$

such that $h(m, n) = \frac{m}{|n|+1}$

$$(m, n) \in \mathbb{Z} \times \mathbb{Z}$$

Solⁿ:-

$$h(1, -3) = \frac{1}{|-3|+1} = \frac{1}{3+1} = \frac{1}{4}$$

$$h(1, 3) = \frac{1}{|3|+1} = \frac{1}{3+1} = \frac{1}{4}$$

$$(1, -3) \neq (1, 3) \Rightarrow h(1, -3) = h(1, 3)$$

$\therefore h$ is not one to one.

Let $b \in \mathbb{Q}$

$b = p/q$ such that $p, q \in \mathbb{Z}$, $q \neq 0$
without loss of generality q be the integer.

$$q > 0 \Rightarrow q-1 \geq 0 \quad |q-1| = q-1$$

Consider $(p, q-1) \in \mathbb{Z} \times \mathbb{Z}$

$$h(p, q-1) = \frac{p}{|q-1|+1} = \frac{p}{q-1+1} = \frac{p}{q} = b$$

$$h(p, q-1) = b$$

$\therefore h$ is onto.

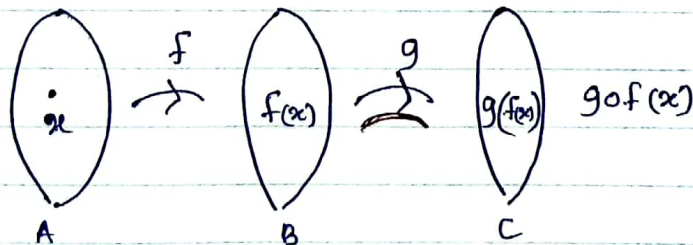
Defⁿ :- Composition map :-

Suppose $f: A \rightarrow B$ and $g: B \rightarrow C$ are function with the property that the Co-domain of f equals to the domain of g . The Composition of f with g ~~is denoted~~ it another function, denote as $g \circ f$ defined as follows.

If $x \in A$, then $g \circ f(x) = g(f(x))$ there for $g \circ f$ sends elements of A to elements of C .

$$\text{i.e. } g \circ f : A \rightarrow C$$

$$f(A) = B$$



Eg:- $f(x) = x+1$ $g(x) = x^2+1$

$$g \circ f(x) = g(f(x))$$

$$= g(x+1)$$

$$= (x+1)^2 + 1$$

$$= x^2 + 2x + 2$$

$$g \circ f(x) = x^2 + 2x + 2$$

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Work
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Th^m :-

Composition of functions is associative

i.e. $f: A \rightarrow B$, $g: B \rightarrow C$, $h: C \rightarrow D$.

Then,

$$(h \circ g) \circ f = h \circ (g \circ f)$$

Th^m :-

Suppose $f: A \rightarrow B$ and $g: B \rightarrow C$. If both f and g are 1-1 then $g \circ f$ is also 1-1.

*** Inverse function :-**

Defⁿ :-

For a set A , the identity function on A is the function $i_A: A \rightarrow A$ defined as

$$i_A(x) = x \quad \forall x \in A.$$

Eg:- $i_{\mathbb{Z}}: \mathbb{Z} \rightarrow \mathbb{Z}$

$$i_{\mathbb{Z}}(n) = n$$

Eg:- $i_{\mathbb{R}}: \mathbb{R} \rightarrow \mathbb{R}$

$$i_{\mathbb{R}}(x) = x$$

Defⁿ:-

given a relation R from A to B the inverse relation of R is the relation from B to A defined as R^{-1}

i.e. ~~If $R = \{(x, y) \in A \times B\}$~~

i.e. If $R = \{(x, y) \mid (x, y) \in A \times B\}$

$$R^{-1} = \{(y, x) \mid (y, x) \in B \times A\}.$$

Eg:- $A = \{a, b\}$, $B = \{1, 2, 3\}$

$$f = \{(a, 2), (b, 3)\}$$

$$f^{-1} = \{(2, a), (3, b)\}$$

th^m:- Let $f: A \rightarrow B$ be a function, then f is bijective if and only if ~~function~~ f^{-1} is a function from $B \rightarrow A$.

Defⁿ:-

If $f: A \rightarrow B$ is bijective then its inverse is the function $f^{-1}: B \rightarrow A$. The functions f and f^{-1} obey the equations ~~$f^{-1} \circ f = i_A$~~ and $f \circ f^{-1} = i_B$.

$$f: A \rightarrow B \quad f^{-1}: B \rightarrow A$$

$$f \circ f^{-1}: A \rightarrow A$$

$$f \circ f^{-1} = i_A$$

$$f^{-1} \circ f: B \rightarrow B$$

$$f^{-1} \circ f = i_B$$

Eg:- $f: \mathbb{R} \rightarrow \mathbb{R}$ such that $f(x) = x^3 + 1$

If f is invertible.

$$f(a) = a^3 + 1$$

$$f(b) = b^3 + 1$$

Solⁿ:- Let $x, y \in \mathbb{R}$

$$\text{Suppose } f(x) = f(y)$$

$$x^3 + 1 = y^3 + 1$$

$$x^3 = y^3$$

$$x = y$$

f is one to one.

Let $b \in \mathbb{R}$ (Co-domain)

$$\text{Suppose } f(x) = b$$

$$x^3 + 1 = b$$

$$x^3 = b - 1$$

$$x = (b-1)^{1/3}$$

$$f((b-1)^{1/3}) = ((b-1)^{1/3})^3 + 1$$

$$= (b-1) + 1$$

$$= b$$

$\therefore f$ is onto

$\therefore f$ is bijective

$\therefore f$ is invertible

f^{-1} exists.

$$\text{Suppose } y = x^3 + 1$$

$$x = (y-1)^{1/3}$$

$$\therefore f^{-1}(x) = (x-1)^{1/3}$$

$$f^{-1}: \mathbb{R} \rightarrow \mathbb{R}$$

$$\begin{aligned}
 f \circ f^{-1}(x) &= f(f^{-1}(x)) \\
 &= f((x-1)^{1/3}) \\
 &= ((x-1)^{1/3})^3 + 1
 \end{aligned}$$

$$= x - 1 + 1$$

$$= x$$

$$f \circ f^{-1}(x) = x = i_k(x)$$

H/w

Eg:- $g: \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z} \times \mathbb{Z}$ $(m, n) \mapsto (m+n, m+2n)$
 $g(m, n) = [m+n, m+2n]$ bijective and find g^{-1} .

Binary operation:-

A binary operation $*$ on a set S is a function.

$$*: S \times S \rightarrow S$$

$$f: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$$

$$*(a, b) = c$$

$$f(a, b) \in \mathbb{R}$$

$$(a, b) \in S \times S, \quad c \in S$$

Eg:- Let $S = \mathbb{R}$ and $*$ be $+$ For $a, b \in \mathbb{R}$

$$a * b = a + b \in \mathbb{R}$$

$$+: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$$

$\therefore +$ is binary operation on $\mathbb{R}$.

$$\cdot(a, b) = a \cdot b \quad a \cdot b = ab \in \mathbb{R}$$

Eg:- Let $S = \mathbb{Z}$ and for any $a, b \in \mathbb{Z}$

$$a * b := \frac{a}{b}$$

$$a * b = \frac{a}{b} \quad a = 2, \quad b = 3$$

$$2 * 3 = \frac{2}{3} \notin \mathbb{Z}$$

This is not a binary operation.

Commutative and associative, binary operations

Defⁿ :-

~~Commutat~~ Let $*: S \times S \rightarrow S$ be binary operation.

- i. If $\forall a, b \in S$
 $a * b = b * a$ we say that $*$ Commutative.
- ii. If $\forall a, b, c \in S$
 $(a * b) * c = a * (b * c)$
 then we say that $*$ associative.

Identity element ;

Dfⁿ :-

Let $*$ be a binary operation on a set S . we say that $e \in S$ is a Identity element for S if

$$\forall a \in S$$

$$e * a = a * e = a$$

eg:- $+: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$

$$a + b = a + b$$

$$a + 0 = a$$

$$0 + b = b$$

$$b + 0 = b$$

$\therefore 0$ is an identity element of $+$ binary operation.

$$\cdot : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$$

$$\forall a, b \in \mathbb{R}$$

$$a \cdot b = ab$$

$$1 \in \mathbb{R}$$

$$\forall a \in \mathbb{R}$$

$$1 \cdot a = a \cdot 1 = a$$

1 is an identity element of $\cdot$ binary operation.

eg:- $* : \mathbb{Q} \times \mathbb{Q} \rightarrow \mathbb{Q}$

$$a * b = \frac{ab}{3}$$

Since $\frac{ab}{3} \in \mathbb{Q}$ $*$ is a binary operation.

$$\forall a \in \mathbb{Q}$$

$$a * b = a$$

$$\frac{ab}{3} = a$$

$$b = 3$$

$$a * 3 = \frac{3a}{3} = a$$

3 is an identity element of $*$

Permutations:-

$$A = \{1, 2, 3, 4\}$$

$$S_4$$

$$|A| = 4 \times 3 \times 2 \times 1$$

$$= 24$$

Defⁿ :-

Let $A = \{a_1, a_2, \dots, a_n\}$ be finite Set with n elements.

The map,

$$a_1 \rightarrow a_2$$

$$a_2 \rightarrow a_3$$

$$\vdots$$

$$a_{n-1} \rightarrow a_n$$

$$a_n \rightarrow a_1$$

$$\begin{pmatrix} a_1 & a_2 & a_3 & \dots & a_{n-1} & a_n \\ a_2 & a_3 & a_4 & \dots & a_n & a_1 \end{pmatrix}$$

Eg:- $A = \{1, 2, 3\}$

$$\sigma_0 = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}$$

$$\sigma_1 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}$$

$$\sigma_2 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}$$

$$\sigma_3 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}$$

$$\sigma_4 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}$$

$$\sigma_5 = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}$$

This permutation Can be written in the cycle form

$$(a_1, a_2, a_3, \dots, a_{n-1}, a_n)$$

Eg :- $A = \{1, 2, 3\}$

$$\sigma_0 = (1), \quad \sigma_1 = (1 \ 2 \ 3), \quad \sigma_2 = (1 \ 3 \ 2)$$

$$\sigma_3 = (1 \ 2)(3) = (1 \ 2) \Rightarrow \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}$$

$$\sigma_4 = (1 \ 3)$$

$$\sigma_5 = (2 \ 3)$$

Length of the cycle $\rightarrow$ (1 2 3) $\rightarrow$ 3

Eg:- (1) - Length 1

(1 2) - Length 2

(1 2 3) - Length 3

$\vdots$

$\vdots$

Note:- Every Permutations Can be written as form Cycle.

Eg:- $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 2 & 4 & 5 & 1 & 6 & 3 & 7 \end{pmatrix}$

$= (1 \ 2 \ 4) (3 \ 5 \ 6)$

* Inverse of a permutation:-

Eg:- $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 5 & 2 & 4 & 1 \end{pmatrix} \xrightarrow{\sigma^{-1}}$

$\sigma = (1 \ 3 \ 2 \ 5) \rightarrow \text{①}$

$\sigma^{-1} = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 3 & 1 & 4 & 2 \end{pmatrix} \rightarrow (1 \ 5 \ 2 \ 3) \rightarrow \text{②}$

$\sigma = (1 \ 3 \ 2 \ 5)$

$\sigma^{-1} = (5 \ 2 \ 3 \ 1) = \text{②}$

Eg:- $\tau = (1 \ 3 \ 5) (7 \ 4 \ 6 \ 8)$

$\tau^{-1} = ((1 \ 3 \ 5) (7 \ 4 \ 6 \ 8))^{-1}$

$\tau^{-1} = (1 \ 3 \ 5)^{-1} (7 \ 4 \ 6 \ 8)^{-1} = (5 \ 3 \ 1) (8 \ 6 \ 4 \ 7)$

~~$\tau^{-1} = (5 \ 3 \ 1) (7 \ 4 \ 6 \ 8)$~~

Eg:- $\tau = (2 \ 3 \ 7) (1 \ 5 \ 6 \ 9)$

$$\tau^{-1} = ((2 \ 3 \ 7) (1 \ 5 \ 6 \ 9))^{-1}$$

$$\tau^{-1} = (2 \ 3 \ 7)^{-1} (1 \ 5 \ 6 \ 9)^{-1}$$

$$\tau^{-1} = (7 \ 3 \ 2) (9 \ 6 \ 5 \ 1)$$

* Product of Permutation is a Composition of the permutation functions:-

Eg:- $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix}$

$$\tau = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix}$$

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 \end{pmatrix}$$

$$\tau = \begin{pmatrix} 2 & 3 & 4 \end{pmatrix}$$

$$\sigma\tau = (1 \ 2 \ 3 \ 4) (2 \ 3 \ 4)$$

$$\sigma(1) = 2$$

$$\tau(1) = 1$$

$$\sigma(2) = 3$$

$$\tau(2) = 3$$

$$\sigma(3) = 4$$

$$\tau(3) = 4$$

$$\sigma(4) = 1$$

$$\tau(4) = 2$$

$$\sigma; A \rightarrow A \quad \tau; A \rightarrow A \quad A = \{1, 2, 3, 4\}$$

$$\sigma \circ \tau(1) = \sigma(\tau(1))$$

$$= \sigma(1)$$

$$\sigma\tau = 2$$

$$\sigma \circ \tau(3) = 1$$

$$\sigma \circ \tau(4) = 3$$

$$\sigma \circ \tau(2) = \sigma(\tau(2))$$

$$= \sigma(3)$$

$$\sigma\tau = 4$$

$$\boxed{\sigma \circ \tau = \sigma\tau}$$

Composition.

$$\sigma\tau = (1 \ 2 \ 4 \ 3)$$

$$\tau\sigma = (2 \ 3 \ 4) (1 \ 2 \ 3 \ 4)$$

$$\tau\sigma(1) = \tau(\sigma(1))$$

$$= 3$$

$$\tau\sigma(3) = 2$$

$$\tau\sigma(2) = \tau(\sigma(2))$$

$$= 4$$

$$\tau\sigma(4) = 1$$

$$\tau\sigma = (1 \ 3 \ 2 \ 4)$$

$$\sigma\tau \neq \tau\sigma$$

$$\sigma^{-1} = (4 \ 3 \ 2 \ 1)$$

$$\tau^{-1} = (4 \ 3 \ 2)$$

$$(1) \ \sigma\sigma^{-1}$$

$$(2) \ \tau\tau^{-1}$$

$$(3) \ \sigma^{-1}\sigma$$

$$(4) \ \tau^{-1}\tau$$

$$(1) \ \sigma\sigma^{-1}$$

$$\sigma\sigma^{-1}(1) = \sigma(\sigma^{-1}(1))$$

$$= \sigma(4)$$

$$\sigma\sigma^{-1}(2) = \sigma(\sigma^{-1}(2))$$

$$= \sigma(1)$$

$$\sigma\sigma^{-1}(3) = \sigma(\sigma^{-1}(3))$$

$$= \sigma(2)$$

$$= 3$$

$$\sigma\sigma^{-1}(4) = \sigma(\sigma^{-1}(4))$$

$$= \sigma(3)$$

$$= 4$$

$$\sigma\sigma^{-1} = (1 \ 2 \ 3 \ 4)$$

$$\sigma\sigma^{-1} = (1)$$

$$(2) \ \tau\tau^{-1}$$

$$\tau\tau^{-1}(1) = \tau(\tau^{-1}(1))$$

$$= \tau(1)$$

$$= 1$$

$$\tau\tau^{-1}(2) = \tau(\tau^{-1}(2))$$

$$= \tau(4)$$

$$= 2$$

$$\tau\tau^{-1}(3) = \tau(\tau^{-1}(3))$$

$$= \tau(2)$$

$$= 3$$

$$\tau\tau^{-1}(4) = \tau(\tau^{-1}(4))$$

$$= \tau(3)$$

$$= 4$$

$$\tau\tau^{-1} = (1)$$

03) $\sigma^{-1}\sigma$

$$\sigma^{-1}\sigma(1) = \sigma^{-1}(\sigma(1))$$

$$= \sigma^{-1}(2)$$

$$= 1$$

$$\sigma^{-1}\sigma(3) = \sigma^{-1}(\sigma(3))$$

$$= \sigma^{-1}(4)$$

$$= 3$$

$$\sigma^{-1}\sigma(2) = \sigma^{-1}(\sigma(2))$$

$$= \sigma^{-1}(3)$$

$$= 2$$

$$\sigma^{-1}\sigma(4) = \sigma^{-1}(\sigma(4))$$

$$= \sigma^{-1}(1)$$

$$= 4$$

$$\sigma^{-1}\sigma = (1)$$

04)

$$\tau^{-1}\tau$$

$$\tau^{-1}\tau(1) = \tau^{-1}(\tau(1))$$

$$= \tau^{-1}(2)$$

$$= 1$$

$$\tau^{-1}\tau(2) = \tau^{-1}(\tau(2))$$

$$= \tau^{-1}(3)$$

$$= 2$$

$$\tau^{-1}\tau(3) = \tau^{-1}(\tau(3))$$

$$= \tau^{-1}(4)$$

$$= 3$$

$$\tau^{-1}\tau(4) = \tau^{-1}(\tau(4))$$

$$= \tau^{-1}(1)$$

$$= 4$$

$$\tau^{-1}\tau = (1)$$

Note:- Every permutation can be written as a product of disjoint cycles.

eg:- $\alpha = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 1 & 5 & 4 \end{pmatrix} = (1 \ 2 \ 3) (4 \ 5)$

order of the Permutation Cycle:-

eg:- $(1 \ 2 \ 3)$ - order 3

(1) - order 1

$(1 \ 2)$ - order 2

* Defⁿ:- A transposition is a permutation which interchanges two elements and leaves every thing else fixed (i.e transposition is a cycle of length 2).

eg:- $(1 \ 2), (3 \ 4)$

Proposition:-

Every Permutation is a product of transposition

$$(a_1 \ a_2 \ a_3 \ \dots \ a_{n-1} \ a_n) = (a_1 \ a_2) (a_1 \ a_3) \dots (a_1 \ a_{n-1}) (a_1 \ a_n)$$

$$(a_1 \ a_2 \ \dots \ a_{n-1} \ a_n) = (a_1 \ a_2) (a_2 \ a_3) \dots (a_{n-1} \ a_n)$$

eg:- $\sigma = (1 \ 2 \ 3 \ 4) = (1 \ 2) (1 \ 3) (1 \ 4)$

$$\sigma = (1 \ 2 \ 3 \ 4) = (1 \ 2) (2 \ 3) (3 \ 4)$$

Product of ^{odd number of} transposition.

$$\tau = (1 \ 2 \ 3 \ 4 \ 5) = (1 \ 2)(2 \ 3)(3 \ 4)(4 \ 5)$$

Product of even number of transpositions.

Note:- Any permutation can be written as a product of even number or odd number of transposition and both.

$$\text{Eg:- } \sigma = (1 \ 3 \ 5 \ 4)(2 \ 6 \ 7)$$

$$= (1 \ 3)(3 \ 5)(5 \ 4)(2 \ 6)(6 \ 7)$$

↑ odd transposition.

$$\sigma = (1 \ 3 \ 5 \ 4)$$

$$O(\sigma) = 4$$

$$\tau = \underbrace{(1 \ 3 \ 5 \ 4)}_4 \underbrace{(6 \ 7 \ 8)}_3$$

$$O(\tau) = \text{LCM (of disjoint cycles)}$$

$$O(\tau) = 12$$

Product of Permutation Cycles.

$$\text{Eg:- } \sigma = (1 \ 2 \ 3 \ 4)(5 \ 6 \ 7)$$

$$\text{Eg:- } \tau = (1 \ 2 \ 3 \ 4)(3 \ 4 \ 7)$$

$$= (1 \ 2 \ 3 \ 4 \ 7 \ 5)$$

$$(1 \ 2 \ 3 \ 4 \ 7 \ 5)$$

$$1 \rightarrow 1$$

$$2 \rightarrow 2$$

$$3 \rightarrow 4 \rightarrow 4$$

$$3 \rightarrow 4$$

Home work :-

$$\text{Eg: } \tau = (1 \ 3 \ 5 \ 7) (4 \ 2 \ 1 \ 3) \\ (1 \ 5 \ 7) (2 \ 3 \ 4)$$

$$(ii) \sigma = (1 \ 2 \ 3 \ 8) (3 \ 6) (7 \ 1) \\ (1 \ 7 \ 2 \ 3 \ 6 \ 8)$$

$$(iii) \sigma = (3 \ 1 \ 7) (1 \ 5 \ 7 \ 2) (6 \ 9 \ 3) \\ (1 \ 5 \ 3 \ 6 \ 9) (2 \ 7)$$

$$\underbrace{(1 \ 2 \ 3)}_3 \underbrace{(4 \ 6 \ 7 \ 5)}_4 = \alpha \quad \text{LCM} = [3, 4] = 12$$

$$\alpha = (1 \ 3) (3 \ 5 \ 7) \\ = (1 \ 3 \ 5 \ 7)$$

$$\alpha^4 = (1)$$

$$\alpha^2 = (1 \ 3 \ 5 \ 7) (1 \ 3 \ 5 \ 7) \\ = (1 \ 5) (2) (3 \ 7) (4) (6)$$

$$\alpha^2 = (1 \ 5) (3 \ 7)$$

$$\alpha^4 = (1 \ 5) (3 \ 7) (1 \ 5) (3 \ 7) (1) (2) (3)$$

$$= (1) (2) (3) (4) (5) (6) (7)$$

$$\alpha^4 = (1)$$

$$\text{Eg :- } \alpha = (1 \ 5 \ 7 \ 2) (3 \ 2 \ 1 \ 4)$$

Find α^{2023}

$$\alpha = (1 \ 5 \ 7 \ 2) (3 \ 2 \ 1 \ 4) = (1 \ 4 \ 3) (2 \ 5 \ 7) \\ (1 \ 4 \ 3) (2 \ 5 \ 7) 3$$

$$\alpha^3 = 1$$

$$\alpha^{2023} = (\alpha^3)^{674} \cdot \alpha \\ = (1)^{674} \cdot \alpha$$

$$= (1) \cdot \alpha = \alpha = (1 \ 4 \ 3) (2 \ 5 \ 7)$$

Atlas

$$\beta = (1 \ 5 \ 3) (2 \ 4 \ 7 \ 6) \quad (1 \ 5 \ 3) (2 \ 4 \ 7 \ 6)$$

$$\beta = (2 \ 4 \ 7 \ 6) (1 \ 5 \ 3)$$

Note:- product of two disjoint cycles are commute.

$\alpha\beta = \beta\alpha$ if α and β are disjoint cycle.

$$\alpha\alpha^{-1} = (1 \ 3 \ 4 \ 7) (7 \ 4 \ 3 \ 1)$$

$$= (1) (2) (3) (4) (5) (6) (7)$$

$$\alpha\alpha^{-1} = (1)$$

Group theory :-

Recall:-

A binary operation $*$ is a function on a set S defined by

$$*: S \times S \rightarrow S$$

$$\forall (a, b) \in S \times S \Rightarrow *(a, b) \in S$$

$$+; \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$$

$$(a, b) \in \mathbb{R} \times \mathbb{R}$$

$$+(a, b) = a + b \in \mathbb{R}$$

$$a * b = a + b$$

$$*(a, b) = a + b - 2$$

$$*: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$$

$$a * b = \frac{ab}{5}$$

* binary operation

$$a * b = \frac{ab}{2}$$

* not binary operation.

Defⁿ :-

A Group is a set G together with a binary operation $*$ defined on G . Such that $(G, *)$

$$(i) \quad a * (b * c) = (a * b) * c \quad \forall a, b, c \in G; \quad (\text{associative property}).$$

$$(ii) \quad \exists \text{ an element } e \in G, \text{ called identity element. Such that } a * e = e * a = a \quad \forall a \in G.$$

$$(iii) \quad \text{For each } a \in G, \exists \text{ an element } a' \in G \text{ such that } a * a' = a' * a = e$$

$\therefore$ Then $(G, *)$ is a Group.

In addition to the above properties.

$$(iv) \quad \forall a, b \in G \quad a * b = b * a \quad (\text{commutative property})$$

Then we say that $(G, *)$ is a Abelian group.

eg:- Let G be a set such that $G \subseteq \mathbb{Q} \setminus \{0\}$ is an operation defined by $\forall a, b \in G$

$$a * b = \frac{ab}{5}$$

Is $(G, *)$ group?

$$* : G \times G \rightarrow G$$

$$\forall a, b \in G$$

$$a * b = \frac{ab}{5} \in G$$

$\Rightarrow *$ is a binary operation

$$i) \forall a, b, c \in G$$

$$a * (b * c) = a * \left(\frac{bc}{5} \right) = \frac{a \left(\frac{bc}{5} \right)}{5} = \frac{abc}{25}$$

$$(a * b) * c = \left(\frac{ab}{5} \right) * c = \frac{\left(\frac{ab}{5} \right) * c}{5} = \frac{abc}{25}$$

$a * (b * c) = (a * b) * c \quad \forall a, b, c \in G$
 associative property hold.

$$ii) \text{ Let } a \in G.$$

Consider

$$a * e = \frac{ae}{5} = a$$

$$e = 5$$

$$(a * 5) = \frac{a5}{5} = a$$

$$5 * a = \frac{5a}{5} = a$$

$\therefore e = 5$ identity exists.

$$iii) \forall a \in G$$

$$a * a' = \frac{aa'}{5} = 5$$

$$a' = \frac{25}{a} \quad \forall a \in G.$$

$$a' * a = \frac{\left(\frac{25}{a}\right)(a)}{5} = 5$$

$$a' * a = a * a' = 5 \quad \forall a \in G$$

$$a' = \frac{25}{a}$$

$\therefore (G, *)$ is a group //

ii). $\forall a, b \in G$

$$a * b = \frac{ab}{5} - \frac{ba}{5} = b * a$$

$\forall a, b \in G, a * b = b * a$

$\therefore (G, *)$ is an Abelian group.

Def :-

A set G with a binary operation $*$ satisfies only associative property, i.e. $(\forall a, b, c \in G, a * (b * c) = (a * b) * c)$ then $(G, *)$ is said to be Semi Group.

Eg:- Let $G = \mathbb{N}$ and $*$ be a operation 'usual addition'.

Is $(G, *)$ Group?

$$a * b = a + b$$

$$*; G \times G \rightarrow G \quad *; \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$$

$$\forall a, b \in G$$

$$*(a, b) = a + b \in \mathbb{N}$$

$\Rightarrow *$ is a binary operation.

Let

~~$$1) \text{ Let } a, b, c \in G$$~~

~~$$a * (b + c) = a * b + a * c$$~~

~~$$a + (b + c) = a + b + c$$~~

~~$$(a + b) + c = a + b + c$$~~

$$(i) \text{ Let } a, b, c \in G$$

$$a * (b * c) = a * (b + c)$$

$$= a + (b + c)$$

$$= (a + b) + c$$

$$= (a * b) + c$$

$$= (a * b) * c$$

 $(G, *)$ is Semi Group.

* Associative Property hold.

$$(ii) \text{ Let } a \in G$$

$$a * e = a$$

$$a + e = a$$

$$e = a - a$$

$$e = 0 \notin G$$

 $\therefore$ identity element in G : $\therefore (G, *)$ not in a Group //Eg:- Let $G = \{1, -1, i, -i\}$ * be a binary operation under usual multiplication?In $(G, *)$ group?(i) Let $1, -1, i \in G$ This is associative. (Home work)

$$(ii) \text{ Let } 1 \in G$$

$$a * e = a$$

$$1 * e = 1$$

$$e = 1$$

$$\text{Let } -1 \in G$$

$$a * e = a$$

$$-1 * e = -1$$

$$e = 1$$

$$\text{Let } i \in G$$

$$a * e = a$$

$$i * e = i$$

$$e = 1$$

$$\sqrt{1} \sqrt{1^2} = 1$$

Let $-i \in G$.

$$a * e = a$$

$$-i * e = -i$$

$$1 * e = 1$$

$\therefore 1$ is an identity element.

$$1 * 1 = 1$$

$$-1 * -1 = 1$$

$$i * -i = -i^2 = 1$$

$$-i * i = -i^2 = 1$$

$$ax^3 + bx^2 + cx + d = 0$$

$$a, b, c, d \in \mathbb{R}$$

$\therefore (G, *)$ is a Group.

$$* \quad ax^2 + bx + c = 0$$

$$\Delta = b^2 - 4ac$$

$$\Delta \geq 0$$

$\Delta = 0 \Rightarrow$ Equal roots

$\Delta > 0 \Rightarrow$ Distinct real roots

$$\Delta < 0$$

$$b^2 - 4ac < 0$$

$$\therefore \sqrt{b^2 - 4ac} = \sqrt{(-1)(b^2 - 4ac)}$$

$$= \sqrt{(-1)} \sqrt{b^2 - 4ac}$$

$$= i \sqrt{b^2 - 4ac}$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-b \pm i \sqrt{b^2 - 4ac}}{2a}$$

$$x_1 = \frac{-b}{2a} + \frac{i \sqrt{b^2 - 4ac}}{2a}; \quad \bar{x}_1 = \frac{-b}{2a} - \frac{i \sqrt{b^2 - 4ac}}{2a}$$

$$z = x + iy$$

$$\bar{z} = x - iy$$

 (*) Let G be the set of all complex n^{th} root of unity.
 i.e. $G = \{z \in \mathbb{C} \mid z^n = 1\}$
 prove that n^{th} root of unity is a group under multiplication.

Solⁿ:-

$$z^n = 1$$

$$z^n - 1 = 0$$

$$(z-1)(z^{n-1} + z^{n-2} + \dots + z + 1) = 0.$$

$$z = 1^{1/n}$$

$$1^{1/n} = 1 + i0 = \cos 0 + i \sin 0$$

$$1^{1/n} = (\cos 0 + i \sin 0)^{1/n}$$

$$= (\cos 2k\pi + i \sin 2k\pi)^{1/n} \quad k = 0, 1, 2, \dots$$

$$= \left(\cos \frac{2k\pi}{n} + i \sin \frac{2k\pi}{n} \right) \quad (k = 0, 1, 2, \dots, n-1) \quad (\text{by De Moivre's } n^{\text{th}})$$

$$k = 0$$

$$\cos 0 + i \sin 0 = 1 + i0 = \cos 0 \left(\frac{2\pi}{n} \right) + i \sin \left(\frac{2\pi}{n} \right) = r^0$$

$$k = 1$$

$$\cos 1 \left(\frac{2\pi}{n} \right) + i \sin 1 \left(\frac{2\pi}{n} \right) = \cos \left(\frac{2\pi}{n} \right) + i \sin \left(\frac{2\pi}{n} \right) = r \quad (\text{say})$$

$$k = 2$$

$$\cos 2 \left(\frac{2\pi}{n} \right) + i \sin 2 \left(\frac{2\pi}{n} \right) = r^2$$

$$k = n-1$$

$$\cos (n-1) \left(\frac{2\pi}{n} \right) + i \sin (n-1) \left(\frac{2\pi}{n} \right) = r^{n-1}$$

$$k = n$$

$$\cos n \left(\frac{2\pi}{n} \right) + i \sin \left(\frac{2\pi}{n} \right) n = \cos 2\pi + i \sin 2\pi = 1 + i0 = r^0$$

$$G = \{r^0, r^1, r^2, \dots, r^{n-1}\}$$

$$G = \left\{ r^i \mid r = \cos\left(\frac{2\pi}{n}\right) + i \sin\left(\frac{2\pi}{n}\right), 0 \leq i \leq n-1 \right\}$$

Let r^i, r^j $0 \leq i < n-1$
 $0 \leq j < n-1$

~~$r^i \cdot r^j$~~ The multiplication

$$\Rightarrow r^i \cdot r^j = r^{i+j}$$

$$i+j > n \mid i+j = n \mid i+j < n$$

$$i+j < n \Rightarrow r^{i+j} \in G$$

$$i+j = n \Rightarrow r^{i+j} \in G$$

$$i+j > n$$

$$n \mid (i+j) \Rightarrow \text{ ~~$r^{i+j} \in G$~~ }$$

$$(i+j) = nk + q \quad \{0 \leq q < n\}$$

$$i+j > n, r^{i+j} = r^{(nk+q)} = (r^n)^k \cdot r^q$$

$$= (r^0)^k \cdot r^q$$

$$= 1^k \cdot r^q$$

$$= r^q, r^q \in G.$$

$$\therefore r^{i+j} \in G.$$

$$G = \{r^0, r^1, \dots, r^{n-1}\} \quad (G, \cdot)$$

$$r^{i+j} \in G \quad \forall i+j$$

$\cdot$ is a binary operation in G .

Axiom 01:-

Let $r^i, r^j, r^k \in G$ $0 \leq i, j, k \leq n-1$

$$\text{ ~~$(r^i \cdot r^j)$~~ \cdot r^k = (r^{i+j}) \cdot r^k$$

$$= r^{i+j} \cdot r^k$$

$$= r^{i+j+k}$$

$$r^{i+j} \in G$$

$$r^{(i+j)+k} \in G.$$

$$\begin{aligned}
 (r^i \cdot r^j) r^k &= r^{i+(j+k)} \\
 &= r^i \cdot r^{j+k} \\
 &= r^i \cdot (r^j \cdot r^k)
 \end{aligned}$$

$\therefore$ associative holds.

(ii) Identity:

$$G = \{r^0, r, \dots, r^{n-1}\}$$

$$= \{1, r, r^2, \dots, r^{n-1}\}$$

$$1 \in G$$

$$1 \cdot r^i = r^i = r^i \cdot 1 \quad 0 \leq i \leq n-1$$

$\therefore 1$ is an identity of G .

(iii) Inverse:

Let $r^i \in G$ be any element of G .

$$r^i \in G, \quad r^{n-i} \in G \quad 0 \leq i \leq n-1$$

$$\begin{aligned}
 r^i \cdot r^{n-i} &= r^{(i+n-i)} \\
 &= r^n = r^0 \\
 &= 1
 \end{aligned}$$

$\therefore$ Inverse exist for every element in G .

$\therefore (G, \cdot)$ is a group //

Eg: Let $G = \{M_{nn} \mid n \times n \text{ matrices with real entries } \det M_{nn} \neq 0\}$.

Take binary operation as matrix multiplication.

Eg:- $G = \{ M_{22} \mid 2 \times 2 \text{ matrices, } \det M_{22} \neq 0 \}$.

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \quad \det A = ad - bc \neq 0 \quad I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$(G, \times)$ is a group.

Prop:-

Let G be a group. Identity of G is unique
pf:-

Suppose e, f be two identity elements of G .

$$e, f \in G$$

Now, Since e is identity.

and $f \in G$

$$ef = f = fe \quad \text{--- ①}$$

Now, f is identity

$$e \in G$$

$$fe = e = ef \quad \text{--- ②}$$

$$\text{①} = \text{②} \Rightarrow \underline{\underline{f = e.}}$$

prop:- Let G be a group. Inverse
w/w of every element in G is unique.

Let $a \in G$

$a', b' \in G$ Such that

$$a'a = aa' = e$$

$$b'a = ab' = e.$$

Def.ⁿ:

Let G be group and $g \in G$. If all the powers g^n ; where $n = 0, 1, 2, \dots$, are the distinct elements of G .

i.e.

$$g^i \neq g^j \quad i \neq j \quad i, j = 0, 1, 2, 3, \dots$$

$$G = \{e, g, g^2, g^3, \dots\}$$

then g is called an element of infinite order.

Def.ⁿ:-

Let G be a group and $a \in G$. Let n be smallest the integer, if it exists, such that $a^n = e$. then ' n ' is called order of a . denoted by $O(a) = n$.

$$a^n = e$$

$$a^n = a \cdot a \cdot a \dots a = e$$

$$+ a^n = a + a + a \dots + a = na$$

$$na = e.$$

$$a^n = e = a * a * a * \dots * a = e$$

$$\uparrow \quad \uparrow$$

binary operation

$$a^n = e$$

$$0, 1, 2, \dots, n-1$$

$$a^1, a^2, \dots, a^{n-1}, a^n = e.$$

Defⁿ :-

A non-empty subset H of a group G is called a subgroup of G if

- a) for $a, b \in H$ implies that $a * b \in H$.
 - b) $e \in H$, where e is an identity element of G .
 - c) $a \in H$ implies that $a^{-1} \in H$.
- we write $H \leq G$ (subgroup of G)

$$\begin{aligned} H &\subseteq G \\ H &\neq \emptyset \\ a, b &\in H \quad ab \in H \end{aligned}$$

$$G = \{z \in \mathbb{C} \mid z^n = 1, n \geq 5\}$$

$$G = \{r^0, r^1, \dots, r^{n-1}\} \quad (n \geq 5)$$

$$H = \{z \in \mathbb{C} \mid z^5 = 1\} \quad H \leq G$$

$$H = \{r^0, r^1, r^2, r^3, r^4\}$$

Subgroup:-

Eg:- Let $G = GL(n, \mathbb{R})$ be the group.

$$G = \{M_{nn} \mid \det M_{nn} \neq 0\}$$

$$H = \{A \in G \mid \det A = 1\}$$

Eg:- $G = \{-1, 1, i, -i\}$

$$\begin{aligned} z^4 &= 1 \\ (z^2 - 1)(z^2 + 1) &= 0 \\ z^2 &= 1 \quad z^2 = -1 \\ z &= \pm 1, \quad z = \pm i \end{aligned}$$

$$\begin{aligned} H &= \{-1, 1\} \quad (i) \quad 1 \in G, \quad 1 \in H \\ (ii) \quad (1)(1) &= 1 \quad (1)(1) = 1 \\ (-1)(1) &= -1 \quad (-1)(-1) = 1 \\ (iii) \quad (-1)(-1) &= 1 \\ (1)(1) &= 1 \end{aligned}$$

$\therefore H \leq G$ $\leq$ - subgroup

Th^m :-

A non-empty subset H of a group G is a subgroup if and only if for any $a, b \in H$ implies $ab^{-1} \in H$.

Pf :-

Let H be a subgroup of G
 $H \leq G$

$\Rightarrow H$ itself a group

Let $a, b \in H$

Since $b \in H \Rightarrow b^{-1} \in H$

$a * b^{-1} = ab^{-1} \in H$

Conversely
 $\Leftarrow$ Suppose $\forall a, b \in H, ab^{-1} \in H$ we must show that $H \leq G$.

$H \neq \emptyset, H \subseteq G$

Since

$a, a \in H$

$\Rightarrow aa^{-1} \in H \subseteq G$

$aa^{-1} = e$

$\Rightarrow e \in H$

ii) Let $a \in H$

$e \in H$

$ea^{-1} \in H$

$a^{-1} \in H$

iii) $\forall a, b \in H$

Since $b \in H, b^{-1} \in H$

$a(b^{-1})^{-1} \in H$

$ab \in H$

binary operation
 first show this

$$\begin{aligned}
 \textcircled{1} \Rightarrow a^{-1}(ab) &\in H \\
 (a^{-1}a)b &\in H \quad H \subseteq G \\
 e b &\in H \\
 b &\in H \quad \# \quad \text{as } b \notin H \\
 &\Rightarrow ab \notin H
 \end{aligned}$$

$$\text{If } ab \in k \text{ — } \textcircled{2} \text{ and } b \in k \therefore b^{-1} \in k \quad (k \leq G)$$

$$\begin{aligned}
 (ab)b^{-1} &\in k \\
 a(bb^{-1}) &\in k \\
 a e &\in k \\
 a &\in k \quad \# \quad \text{as } a \notin k \\
 \Rightarrow ab &\notin k.
 \end{aligned}$$

$$\begin{aligned}
 \Rightarrow ab &\notin H \cap k \quad \# \\
 ab &\notin H \cup k \quad \# \quad H \cup k \leq G
 \end{aligned}$$

Our Assumption is wrong.
 $\Rightarrow k \subseteq H$ or $H \subseteq k$.

Eg:- H and k Subgroups of G . Is $H \cap k$ Subgroup of G ?
 Justify your answer.

n/w:- Properties of Groups;

i) The identity element in a group G is unique.

ii) In a group G the inverse element is unique.

iii) In a group G ,

$$a) \quad (a^{-1})^{-1} = a \quad \forall a \in G.$$

$$b) \quad (ab)^{-1} = b^{-1}a^{-1} \quad \forall a, b \in G$$

$$(c) \quad e^{-1} = e$$

iv) In a group G

a) If $ax = ay \Rightarrow x = y \quad \forall a, x, y \in G$

b) If $xa = ya \Rightarrow x = y$

v) In G given $a, b \in G$.

a) then $\exists$ a unique element $x \in G$ such that $ax = b \Rightarrow x = a^{-1}b$ ($x = \underbrace{b a^{-1}}_b$)

b) $\exists$ a unique element $y \in G$ such that $ya = b \Rightarrow y = ba^{-1}$

vi) If $a^{-1} = a \quad \forall a \in G$ then G is abelian.

Cyclic group:-

Defⁿ:-

A group G is called Cyclic if for some $a \in G$ every element $x \in G$ is of the form a^n , where n is some integer.

$$x = a^n \quad n \in \mathbb{Z}$$

$$G = \{r^0, r^1, r^2, \dots, r^{n-1}\}$$

$= \langle r \rangle$ - generator

$$a \in G, \quad a^n = \underbrace{a * a * \dots * a}_{n \text{ times}}$$

Then the element a is called the generator of G .

$$G = \{x \mid x = a^n, n \in \mathbb{Z}, a \in G\}$$

Note:-

1. The cyclic group generated by a is denoted by $\langle a \rangle$

2. If the binary operation is addition.
 $G = \langle a \rangle = \{x \mid x = na, n \in \mathbb{Z}\}.$

03. The Cyclic group may have more than one generator.

Eg:- Fourth root of unity.

$$G = \{z \in \mathbb{C} \mid z^4 = 1\}$$

$$= \{-1, 1, i, -i\}$$

$$i \in G$$

$$i = i$$

$$G = \langle i \rangle$$

$$i \cdot i = i^2 = -1$$

G is Cyclic group

$$i \cdot i \cdot i = i^3 = -i$$

$$G = \langle -i \rangle.$$

$$i \cdot i \cdot i \cdot i = i^4 = 1$$

Eg:- $\mathbb{Z}_3 = \{0, 1, 2\} \Rightarrow (\mathbb{Z}_3, +)$ is Cyclic group.

$$\mathbb{Z}_p = \{0, 1, 2, 3, \dots, p-1\}$$

th^m:-

A Cyclic group of order n has $\phi(n)$ generators.

$\phi(n)$ - Euler ϕ function.

$$n = p_1^{r_1} p_2^{r_2} \dots p_k^{r_k}$$

$p_1, p_2, \dots, p_k$ are distinct prime numbers

$$r_1, r_2, \dots, r_k \in \mathbb{N}$$

$$n = 24 = 2^3 3^1$$

$$\phi(n) = n \left(1 - \frac{1}{p_1}\right) \left(1 - \frac{1}{p_2}\right) \cdots \left(1 - \frac{1}{p_k}\right)$$

$$\text{If } n = p_1^{r_1} \cdots p_k^{r_k},$$

$$n = 60 = 2^2 \cdot 3^1 \cdot 5^1$$

$$\phi(60) = 60 \left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{3}\right) \left(1 - \frac{1}{5}\right)$$

$$= 60 \left(\frac{1}{2}\right) \left(\frac{2}{3}\right) \left(\frac{4}{5}\right)$$

$$= 60 \times \frac{4}{15}$$

$$= 16$$

$$n = 4 = 2^2$$

$$\phi(4) = 4 \left(1 - \frac{1}{2}\right)$$

$$= 4 \times \frac{1}{2}$$

$$= 2$$

Th^m: Every Cyclic group is abelian group.

Chapter 04 :-

Introduction to Number theory:

$\mathbb{Z}$ - Set of integer

$\mathbb{N}$ - Set of natural number

$\mathbb{P}$ - Set of Prime number.



Well-ordering principle:-

Statement,

every non-empty set of non negative integers contain a least (or smallest) element.

* Archimedean property:-

if a and b are any positive integers
then $\exists$ a the integer n such that $na \geq b$

* Mathematical Induction:-

① First principle of finite Induction.

Let S be a set of all positive integers.
with the following properties.

- i. Integer $1 \in S$
- ii. whenever the integer $k \in S$, the next integer $k+1 \in S$
Then S is the set of all positive integers.

$$S = \{1, 2, 3, \dots, p, p+1, \dots, n\}$$

pf:-

Let T be the set of positive integer which are not in S .

Assume that $T \neq \emptyset$

$$\therefore T \subseteq \mathbb{Z}^+$$

by well-ordering principle T has least element say 'a'

$$\text{Since } 1 \in S \Rightarrow 1 \notin T$$

$$\Rightarrow a > 1$$

$$a-1 > 0$$

$$0 < a-1 < a$$

and $a-1 \notin T$ ($\because$ a is smallest element in T)

$$\Rightarrow a-1 \in S$$

$$\text{Let } k = a-1 \in S$$

by (ii) then $k+1 \in S$

$$(a-1)+1 \in S$$

$$a \in S$$

$$\therefore T = \emptyset$$

$$\Rightarrow S = \mathbb{Z}^+$$

Division Algorithm:-

Th^m:-

given integers ~~a~~ a and b , $b > 0$. Then $\exists$ unique integers q and r satisfying

$$a = qb + r ; \quad 0 \leq r < b$$

The integer q and r ~~any~~ are called the quotient and remainder in the division of ~~by~~ a by b .

Pf.:

we start the proof by defining the set $S = \{a - xb \mid x \in \mathbb{Z}, a - xb \geq 0\}$ is non empty.

So, $a \in \mathbb{Z}$ $|a| \geq 0$

Let $x = -|a|$

Now Consider,

$$\begin{aligned} a - xb & \text{ Such that} \\ &= a - (-|a|)b \\ &= a + |a|b. \end{aligned}$$

Since $b \geq 1$

$$\begin{aligned} a - xb &= a + |a|b \\ &\geq a + |a| \\ &\geq 0 \end{aligned}$$

$$a - xb \geq 0$$

$$a - xb \in S \text{ as } x = -|a|$$

$$\therefore S \neq \emptyset$$

As S is non-empty by well ordering principle S contains a smallest integer Call ' r ', $r \geq 0, r \in S$

$$r \in S \Rightarrow \exists q \in \mathbb{Z} \text{ i.e. } x = q$$

$$r = a - bq ; r \geq 0.$$

Suppose $r \geq b$

$$a - bq \geq b$$

$$a - b(q+1) \geq 0$$

$\Rightarrow a - b(q+1) \in S \neq$ as $a - bq$ is smallest.

$$r \not\geq b \Rightarrow r < b$$

$$\therefore a = bq + r ; 0 \leq r < b.$$

Suppose uniqueness of q and r .

Suppose q', q and r', r are such that,

$$a = bq + r \quad 0 \leq r < b$$

$$a = bq' + r' \quad 0 \leq r' < b; \quad r' \geq r$$

$$\Rightarrow bq + r = bq' + r'$$

$$b(q - q') = r' - r \quad (r' - r \geq 0)$$

$$b|q - q'| = |r' - r| \quad (*) \quad |r' - r| = r' - r$$

$$\Rightarrow |r' - r| < b$$

$$(*) \Rightarrow b|q - q'| < b$$

$$\Rightarrow |q - q'| < 1$$

$$0 \leq r < b \quad -b < -r \leq 0$$

$$0 \leq r' < b \quad 0 \leq r' < b$$

$$-b + 0 < r' - r < b + 0$$

$$-b < r' - r < b$$

~~Since $q, q' \in \mathbb{Z}$~~

$$0 \leq |q - q'| < 1$$

Since $q, q' \in \mathbb{Z}$

$$\Rightarrow |q - q'| = 0$$

$$\Rightarrow q = q'$$

$$(*) \quad (r' - r) = 0$$

$$\Rightarrow r' = r$$

$\therefore$ uniqueness holds //

Corollary:-

If a and b are integers with $b \neq 0$ then $\exists$ unique integer q and r such that $a = bq + r$, $0 \leq r < |b|$

Eg:- Square of an integer leaves, the remainder 0 or 1 upon division by 4.

$$n \in \mathbb{Z}$$

$$n^2 \in \mathbb{Z}^+$$

$$4 \mid n^2$$

$$n^2 = 4q + r \quad ; \quad r = 0 \text{ or } r = 1$$

Let $n \in \mathbb{Z}$:

$$n = 2k$$

or

$$n = 2k+1$$

$$k \in \mathbb{Z}$$

$$k \in \mathbb{Z}$$

$$n^2 = 4k^2$$

$$n^2 = (2k+1)^2$$

$$4 \mid n^2$$

$$= 4k^2 + 4k + 1$$

$$\text{remainder} = 0$$

$$= 4(k^2 + k) + 1$$

$$4 \nmid n^2 \Rightarrow \text{remainder} = 1$$

If n^2 divides by 4 then remainder is 1

Properties:-

For integers a, b, c following holds.

01. $a \mid 0$, $1 \mid a$, $a \mid a$ $\forall a$

02. $a \mid 1$ iff $a = \pm 1$

03. if $a \mid b$, $c \mid d$ then $ac \mid bd$.

04. if $a \mid b$, $b \mid c$ then $a \mid c$

05. $a \mid b$, $b \mid a$ iff $a = \pm b$.

06. $a \mid b$ and $b \neq 0$ then $|a| \leq |b|$

Defⁿ:-

Let a and b be given integers, with at least one of them non-zero, the greatest Common division (gcd) of a and b , denoted by $\gcd(a, b)$ (or (a, b)) is the positive integer G , satisfying

- i. $d|a$ and $d|b$
- ii. If $c|a$ and $c|b$ then $c \leq d$.

Eg:- $\gcd(-12, 30) = \underline{\underline{6}}$

* Th^m:- Given integers a and b not both of which are zero. $\exists$ integers x and y such that $\gcd(a, b) = ax + by$.

Pf:-

Consider the set S of all the linear combinations of a and b .

$$S = \{ au + bv \mid au + bv > 0, u, v \in \mathbb{Z} \}.$$

Note that: $S \neq \emptyset$; if $a \neq 0$ then $|a| > 0$,

$$|a| = au + bv > 0$$

$$\left. \begin{array}{l} a < 0; u = -1 \\ a > 0; u = 1 \end{array} \right\} au + bv \in S$$

$$\therefore S \neq \emptyset$$

Since S is non-empty by well ordering-Principle S has smallest element say ' d ' $d \in S$, $d > 0$

$$\Rightarrow d = ax + by, \quad x, y \in \mathbb{Z}$$

Claim: $d = \gcd(a, b)$

using division algorithm

we can obtain q and r such that,

$$a = qd + r \quad 0 \leq r < d.$$

$$r = a - qd$$

$$= a - q(ax + by)$$

$$= a(1 - qx) + b(-qy)$$

$$r = ax' + by' \quad x', y' \in \mathbb{Z}$$

$$x' = 1 - qx$$

$$y' = -qy.$$

If $r > 0$ $ax' + by' > 0$

$\Rightarrow r \in S$ but $r < d$

Since d is smallest $r \notin S$ #

$$r \neq 0$$

$$\Rightarrow r = 0$$

$$\Rightarrow a = qd \Rightarrow d | a$$

$$//^{\circ} d | b$$

$\Rightarrow d$ is common divisor of a and b

Now if c and divisor of a and b :

$$\Rightarrow c | ax + by \quad c | a, c | b$$

$$c | d$$

$$c = |c| \leq |d| = d$$

$$c \leq d$$

$$d = \gcd(a, b)$$

Relative Prime:-

$$(a, b) = 1$$

Then a and b are relatively Prime.

Eg:- $(3, 8) = 1$

Euclidean Algorithm.

a, b two integers such that $a \geq b > 0$

$$\gcd(|a|, |b|) = \gcd(a, b)$$

First Step. Apply division Algorithm.

$$a = qb + r, \quad 0 \leq r < b.$$

If $r_1 = 0$, $b | a$.

Step II:-

If $r_1 \neq 0$

Then divide b by r_1 .

$$\Rightarrow b = q_2 r_1 + r_2, \quad 0 \leq r_2 < r_1$$

If $r_2 = 0$, $r_1 | b$.

If $r_2 \neq 0$ repeat the same process.

$$r_1 = q_3 r_2 + r_3, \quad 0 \leq r_3 < r_2$$

$$r_{n-2} = q_n r_{n-1} + r_n, \quad 0 \leq r_n < r_{n-1}$$

$$r_{n-1} = q_{n+1} r_n + 0.$$

Eg:- $(78, 31) = 1$ ←

$$a = 78, b = 31$$

$$78 = 2(31) + 16$$

$$31 = 1(16) + 15$$

$$16 = 1(15) + \textcircled{1}$$

$$15 = 1(15) + 0$$

Eg:- $(12378, 3054) = 6$ ←

$$a = 12378, b = 3054$$

$$12378 = 4(3054) + 162$$

$$3054 = 18(162) + 138$$

$$162 = 1(138) + 24$$

$$138 = 5(24) + 18$$

$$24 = 1(18) + \textcircled{6}$$

$$18 = 3(6) + 0$$

If $\gcd(a, b) = d$

$$d = ax + by \quad x, y \in \mathbb{Z}$$

$$24 = 1(18) + 6$$

$$6 = 24 - 1(18)$$

$$= 24 - 1(138 - 5(24))$$

$$= 6(24) - 1(138)$$

$$= 6(162 - 1(138)) - 1(138)$$

$$= 6(162) - 7(138)$$

$$= 6(162) - 7(3054 - 18(162))$$

$$= 132(162) - 7(3054)$$

$$= 132(12378 - 4(3054)) - 7(3054)$$

$$6 = 132(12378) - 535(3054)$$

$$6 = (12378)(132) + (3054)(-535)$$

$$x = 132, y = -535$$

**** theorem: Linear Diophantine equation.**

The linear diophantine equation $ax+by=c$ has a solution iff $d|c$ where $d = \gcd(a,b)$. If x_0, y_0 is any particular solution of this equation, then all other solutions are given by

$$x = x_0 + (b/d)t, \quad y = y_0 - (a/d)t, \quad t \in \mathbb{Z}$$

Pf :- Suppose $ax+by=c$ has solution. Say x_0, y_0 .

$$ax_0 + by_0 = c \quad \text{--- (1)}$$

Let $d = \gcd(a, b)$

$$d|a \quad \text{and} \quad d|b$$

$$\Rightarrow a = dr, \quad r \in \mathbb{Z}, \quad b = ds, \quad s \in \mathbb{Z}$$

$$\text{(1)} \Rightarrow c = ax_0 + by_0$$

$$c = drx_0 + dsy_0$$

$$c = d(\underbrace{rx_0 + sy_0}_{\in \mathbb{Z}})$$

$$\Rightarrow d|c$$

Conversely, Suppose $d|c$.

$$d = \gcd(a, b)$$

To show that $ax+by=c$ has a solution

As $d|c \Rightarrow c = dt$ for some $t \in \mathbb{Z}$.

Since $d = \gcd(a, b)$

$$\Rightarrow d = ax + by \quad x, y \in \mathbb{Z}$$

Let $x_0, y_0 \in \mathbb{Z}$

$$d = ax_0 + by_0$$

$$dt = a(x_0 + t) + b(y_0 + t)$$

$$c = a(x_0 + t) + b(y_0 + t)$$

$$\Rightarrow c = ax' + by'$$

$$x' = x_0 + t$$

$$y' = y_0 + t$$

$$24 = \underbrace{(2 \times 2 \times 2)}_{2 \times 2 \times 2 = 8} \times 3 = 8 \times 3 \quad \therefore \gcd(24, 16)$$

$$16 = \underbrace{(2 \times 2 \times 2)}_{2 \times 2 \times 2 = 8} \times 2 = 8 \times 2 = 8$$

$\Rightarrow$ ① has a Solution x', y'

2nd part :-

To prove the 2nd part,
If x_0, y_0 is any particular solution of $ax + by = c$
 $ax + by = c$

$$\Rightarrow ax_0 + by_0 = c \quad \text{--- ②}$$

Let x', y' be any solution $ax' + by' = c \quad \text{--- ③}$

$$\begin{aligned} \text{②} = \text{③} \Rightarrow ax_0 + by_0 &= ax' + by' \\ a(x_0 - x') &= b(y' - y_0) \quad \text{--- ④} \\ d &= \gcd(a, b) \\ d|a, d|b \end{aligned}$$

$$a = dr \quad b = ds$$

$$\Rightarrow (r, s) = 1$$

$$\begin{aligned} \text{④} \quad dr(x_0 - x') &= ds(y' - y_0) \\ r(x_0 - x') &= s(y' - y_0) \end{aligned}$$

$$\Rightarrow r | s(y' - y_0) \quad \left\{ \begin{array}{l} s | (x_0 - x')r \end{array} \right.$$

$$r \nmid s \Rightarrow r | (y' - y_0) \quad \left\{ \begin{array}{l} s \nmid r \Rightarrow s | (x_0 - x') \end{array} \right.$$

$$\Rightarrow y' - y_0 = rt \quad \text{for some } t \in \mathbb{Z} \quad \left\{ \begin{array}{l} x_0 - x' = st \quad \text{for some } t \in \mathbb{Z} \end{array} \right.$$

$$x_0 - x' = st$$

$$a = dr$$

$$x' = x_0 - st$$

$$r = a/d$$

$$x' = x_0 - \left(\frac{b}{d}\right)t$$

$$b = ds$$

$$s = b/d$$

$$y' = y_0 + \left(\frac{a}{d}\right)t$$

$$t \rightarrow -t$$

$$x' = x_0 + \left(\frac{b}{d}\right)t$$

$$y' = y_0 - \left(\frac{a}{d}\right)t$$

eg:- $172x + 20y = 1000$

Find the general solution.

$$a = 172$$

$$b = 20$$

$$(a, b) = 4$$

$$4 \mid 1000$$

$\Rightarrow$ this has solution

$$x_0 = 5, y_0 = 7$$

$$\frac{a}{d} = \frac{172}{4} = 43$$

$$\frac{b}{d} = \frac{20}{4} = 5.$$

$$x' = 5 + 5t \quad t \in \mathbb{Z}.$$

$$y' = 7 - 43t \quad t \in \mathbb{Z}.$$

$$t = 1, \quad x' = 10$$

$$y' = -36.$$